Hyeongchan Ham

M.S. Researcher, KAIST

hyeongchan.ham@kaist.ac.kr | hyeongchanham.github.io | LinkedIn

Research Interests

Practical Safety for Learning-Enabled Robots • **Safe Robot Learning** • **Semantic Safety** • **Learning-Based Control**

Developing safety methods that remain applicable to modern learning-based robotic systems with neural perception and learned dynamics, while extending beyond collision avoidance to semantic notions of safety.

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea
M.S. in Electrical Engineering; Advisor: Prof. Heejin Ahn — *Feb. 2025 – Expected Feb. 2027*

Ulsan National Institute of Science and Technology (UNIST) — Ulsan, Republic of Korea
B.S. in Electrical and Computer Engineering — *Feb. 2017 – Feb. 2021*
GPA: 4.01/4.30 (97.10/100); **Summa Cum Laude**

Ongoing Research

- **Language-Conditioned Semantic Safety for Robot Navigation** — Extending S2-NAMO into the latent space of a vision-language model (VLM), where language-provided context conditions the semantic safety specification so that the same scene can induce different safety assessments and planning behavior depending on task context.
- **Semantically Safe Global Planning with 3DGS** — Extending S2-NAMO toward global navigation by combining visual/semantic risk estimates from 3DGS scene representations with sampling-based global planning (RRT/RRT*) so that the planner accounts for semantic failure modes beyond geometric collision.

Publications

International Journal Publications

[J1] **DRO-EDL-MPC: Evidential Deep Learning-Based Distributionally Robust Model Predictive Control for Safe Autonomous Driving**
Hyeongchan Ham, Heejin Ahn
IEEE Robotics and Automation Letters (RA-L), 11(3):3015–3022, 2026.
Presented at **IEEE International Conference on Robotics and Automation (ICRA 2026)**.
Received the **Outstanding Poster Award at BIEN 2025**.
Presented at the **CoRL 2025 SAFE-ROL Workshop**.

[J2] **Multi-Class Multi-Object Tracking in Aerial Images Using Uncertainty Estimation**
Hyeongchan Ham, Junwon Seo, Junhee Kim, Chungsu Jang
Korean Journal of Remote Sensing, 40(1):115–122, 2024. *(English)*
Indexed in **Scopus, Web of Science ESCI, and KCI**.

International Conference Publications

[C1] **S2-NAMO: Semantically Safe Navigation Among Movable Obstacles**
Hyeongchan Ham, Boseok Kim, Minwoo Song, Heejin Ahn
RSS 2026 Workshop on Rethinking What It Means to be "Safe" for Generalist Robots — **Spotlight**, 2026.
Poster presentation at **Nature Conference: Autonomous Robotics**, Daejeon, Korea, Oct. 2026.

[C2] **Probabilistic Recursively Feasible Motion Planning Under Uncertain Environments**
Hyeontae Sung, **Hyeongchan Ham**, Junyoung Park, Kai Ren, Heejin Ahn
23rd IFAC World Congress, Busan, Republic of Korea, 2026.

[C3] **Safe Path Planning with Visual Information in Gaussian Splatting Maps**
Hyeongchan Ham, Heejin Ahn
13th International Conference on Robot Intelligence Technology and Applications (RiTA) — **Oral**, London, UK, 2025.

Domestic Publications

[D1] **VLM-Based Obstacle Breakthrough Decision-Making for Autonomous Vehicles**
Wonjune Seo et al. (including **Hyeongchan Ham**)
6th Korea Artificial Intelligence Conference, pp. 259–260, 2025.
Best Paper Award.

[D2] **A Survey on Safe Model Predictive Control and Reinforcement Learning**
Hyeongchan Ham, Heejin Ahn
Journal of Institute of Control, Robotics and Systems, 31(2):87–97, 2025.
Indexed in **Scopus and KCI**.

[D3] **Unsupervised Learning with Pseudo-Labels for Object Detection in LiDAR Point Clouds**
Hyeongchan Ham, Minsik Jeon, Sukwon Jang
Korea Military Science and Technology Society Fall Conference, 2023.

[D4] **LiDAR Auto-Labeling System using Multi-Agent Perception**
Hyeongchan Ham, Sukwon Jang, Insu Hwang
19th Korea Robotics Society Annual Conference (KRoC), 2024.

Research Experience

KAIST Control and Intelligent Systems Laboratory (CIS Lab) — Daejeon, Republic of Korea
Graduate Researcher; Advisor: Prof. Heejin Ahn — *Jun. 2024 – Present*

- **Independent Research**
  Semantic safety for learning-based navigation — Developed latent-space semantic safety methods for NAMO by combining learned visual world representations with Hamilton–Jacobi reachability; resulted in **S2-NAMO [C1]**, an RSS 2026 Workshop Spotlight.
- **ADD-Sponsored Research Project**
  VLA navigation — Developing safety-aware VLA navigation methods that incorporate semantic safety considerations into modern vision-language-action policies for robot navigation.
- **Independent Research**
  Safe control under perception uncertainty — Developed evidential uncertainty-aware distributionally robust MPC for autonomous driving, adapting control conservativeness to perception confidence; resulted in **DRO-EDL-MPC [J1]**, IEEE RA-L / ICRA 2026.
- **Samsung Electronics-Sponsored Research Project**
  PAC-Bayes active learning for label-efficient model training — Developed a PAC-Bayesian active-learning method with pseudo-labeling to estimate the minimum labeling budget needed to reach target model performance before costly annotation, and validated the approach on semiconductor process data.

Agency for Defense Development (ADD) – AI Autonomy Center Daejeon, Republic of Korea
Researcher, assigned through the Research Officers for National Defense (ROND) Program *Jun. 2021 – May 2024*

- **AI-enabled UAV swarm operations** — Developed real-time small-object detection and tracking models for multi-UAV systems. Compressed and optimized the models for deployment on **edge devices** and validated them on the target onboard runtime. Designed cooperative-perception components for sharing target-recognition information across multiple agents; resulted in **[J2]**.
- **Adaptive path planning and situational awareness** — Built a ROS2-based LiDAR perception stack and 3D object detection pipeline for autonomous UGVs. Addressed domain shift from public urban datasets to off-road environments using pseudo-label-based learning and multi-agent auto-labeling **[D3, D4]**. Integrated the autonomy software with a **full-scale UGV platform** and conducted deployment and field experiments in **off-road environments**.

OMNIA – UNIST System Laboratory Ulsan, Republic of Korea
Research Intern; Advisor: Prof. Myeongjae Jeon *Jul. 2020 – Feb. 2021*

- **Continual learning** — Developed an efficient replay-buffer data-loading system focused on maximizing replay-stream throughput for continual learning experiments.

Work Experience

Research Officers for National Defense (ROND) Program Republic of Korea
Research Officer, Republic of Korea Air Force *Jun. 2021 – May 2024*

- Selected as one of only **20 outstanding STEM university students nationwide** for the ROND program, a highly selective national program supporting full-time science and technology research for national defense.
- Served as a commissioned officer in the **Republic of Korea Air Force** and completed military service at the rank of **First Lieutenant**.
- Received the **Minister of National Defense Commendation** for exemplary military service and research contributions as a Research Officer for National Defense.

Honors & Awards

Next-Generation Engineering Researcher 2026
Institute for Promotion of Engineering and Science of Korea (IPESK).

Excellence Award, miliTECH Challenge Dec. 2025
Awarded by the Superintendent of the Republic of Korea Air Force Academy.

Best Paper Award, 6th Korea Artificial Intelligence Conference Sep. 2025
Awarded for **[D1]**.

Outstanding Poster Award, BIEN 2025 Aug. 2025
Awarded for **[J1]**.

Minister of National Defense Commendation, Republic of Korea 2024
Awarded for exemplary service and research contributions as a Research Officer for National Defense.

Scholarships & Research Grants

Hyundai Motor Chung Mong-Koo Scholarship (Future Industries) 2025 – Present
KRW 7.0M/semester (approx. US$5.2K/semester).

Master's Research Grant, National Research Foundation of Korea (NRF) 2025 – 2026
Research Subsidies for Master's Students; KRW 12.0M (approx. US$8.9K) for one year.

Conference Travel Grant for RSS 2026 2026
Hyundai Motor Chung Mong-Koo Foundation; up to KRW 3.0M (approx. US$2.2K).

Global Excellence Scholarship 2026
Hyundai Motor Chung Mong-Koo Foundation; KRW 2.0M (approx. US$1.5K).

Korea National Scholarship of Excellence (Science and Engineering) 2019 – 2020
KRW 2.5M/semester (approx. US$1.8K/semester).

UNIST Academic Performance Scholarship 2017 – 2020
Full tuition.

Selected Talks & Presentations

Invited Talk, 41st ICROS Annual Conference Jul. 2026
Institute of Control, Robotics and Systems (ICROS); Special Program: Hot Papers to Read Now; presentation on **[J1]**.

Research Presentation, Hyundai Motor Chung Mong-Koo Foundation Apr. 2026
Science & Technology Academic Exchange Program; presentation on **[J1]**.

Research Presentation, Hyundai Motor Chung Mong-Koo Foundation Sep. 2026
Science & Technology Academic Exchange Program; presentation on **[C1]**.

Teaching & Mentoring Experience

Teaching Assistant, Signals and Systems, KAIST Fall 2026
Course conducted in English.

Teaching Assistant, Data Structures, UNIST Fall 2020
Course conducted in English.

Mentor, CSE Winter School, UNIST Winter 2020–2021
Linux Tutoring Program; conducted in Korean.

Research Mentor, miliTECH Challenge at KAIST 2025
Mentored the VLM-based obstacle-breakthrough project.
The follow-on study **[D1]** received the **Best Paper Award** at the 6th Korea Artificial Intelligence Conference.

Technical Skills

Programming: Python, C/C++, Java

Robotics/ML: PyTorch, TensorFlow, JAX, ROS2, Isaac Sim/Lab, CARLA

Deployment/Tools: Docker, Linux, ONNX, TensorRT, DeepStream, MLflow